

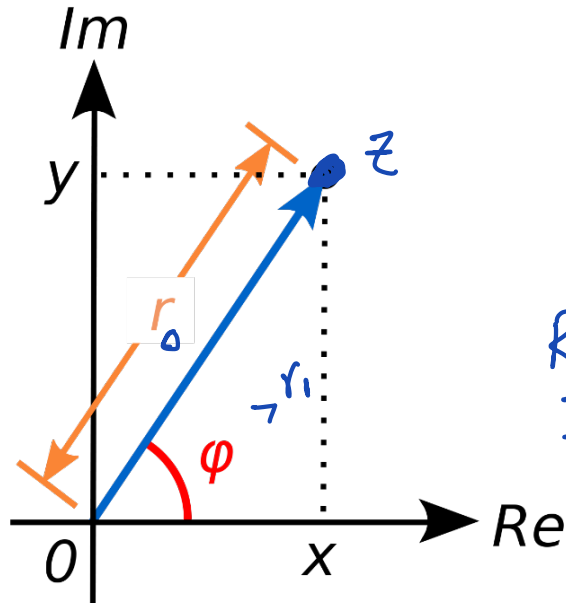
L#24: Complex numbers

$$x^2 - 1 = 0.$$

$$x^2 \geq 0, \quad \underline{x^2 = -1}$$

$$\underline{j^2 = -1}$$

$$x^2 + 1 = 0$$



$$z = \underline{x} + j \underline{y}$$

$$\operatorname{Re}(z) = x.$$

$$\operatorname{Im}(z) = j \underline{y}$$

Rectangular form $z = x + jy$, $x, y \in \mathbb{R}$. j is imaginary unit

$$r = \sqrt{x^2 + y^2}. \quad \underline{x = r \cos \varphi}, \quad \underline{y = r \sin \varphi}$$

$$\underline{z = r \cos \varphi + jr \sin \varphi.}$$

Polar form $\underline{z = r \angle \varphi}$

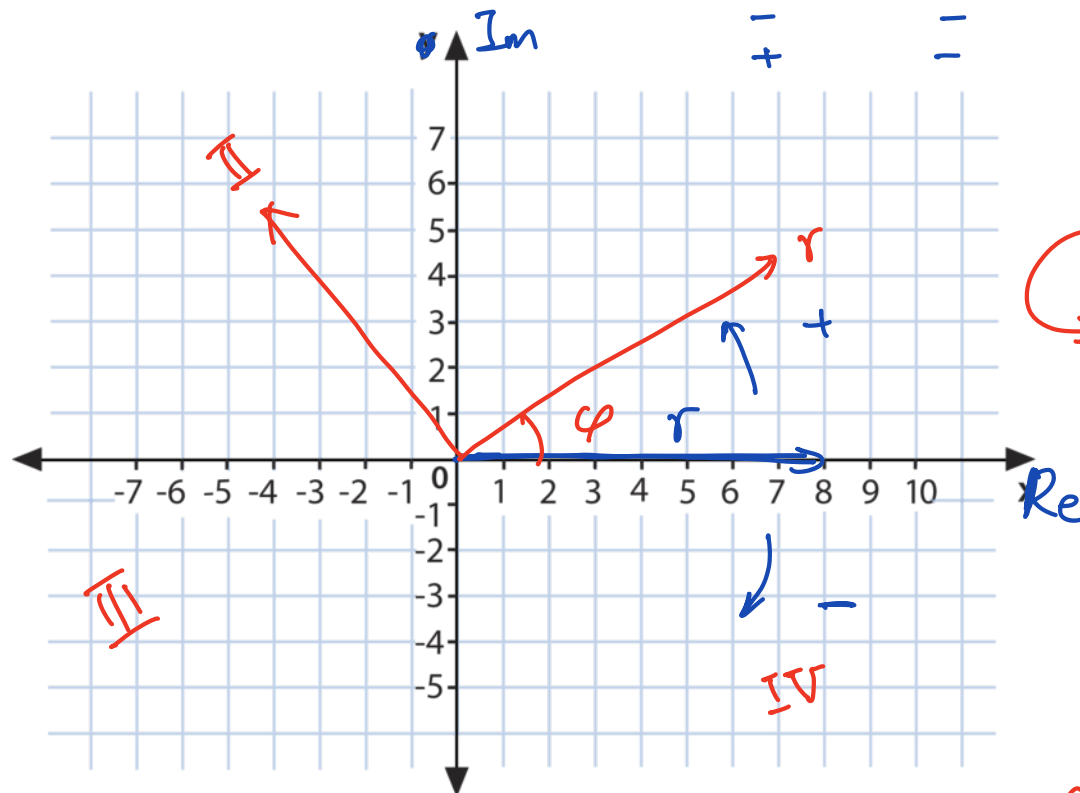
$$r = \sqrt{x^2 + y^2}$$

$$\tan \varphi = \frac{y}{x}, \quad \varphi = \tan^{-1}\left(\frac{y}{x}\right).$$

Exponential form $z = r \cos \varphi + jr \sin \varphi.$
 $= r (\cos \varphi + j \sin \varphi).$

$$\boxed{e^{j\varphi} = \cos \varphi + j \sin \varphi} = \underline{\underline{r e^{j\varphi}}}$$

Understand angles

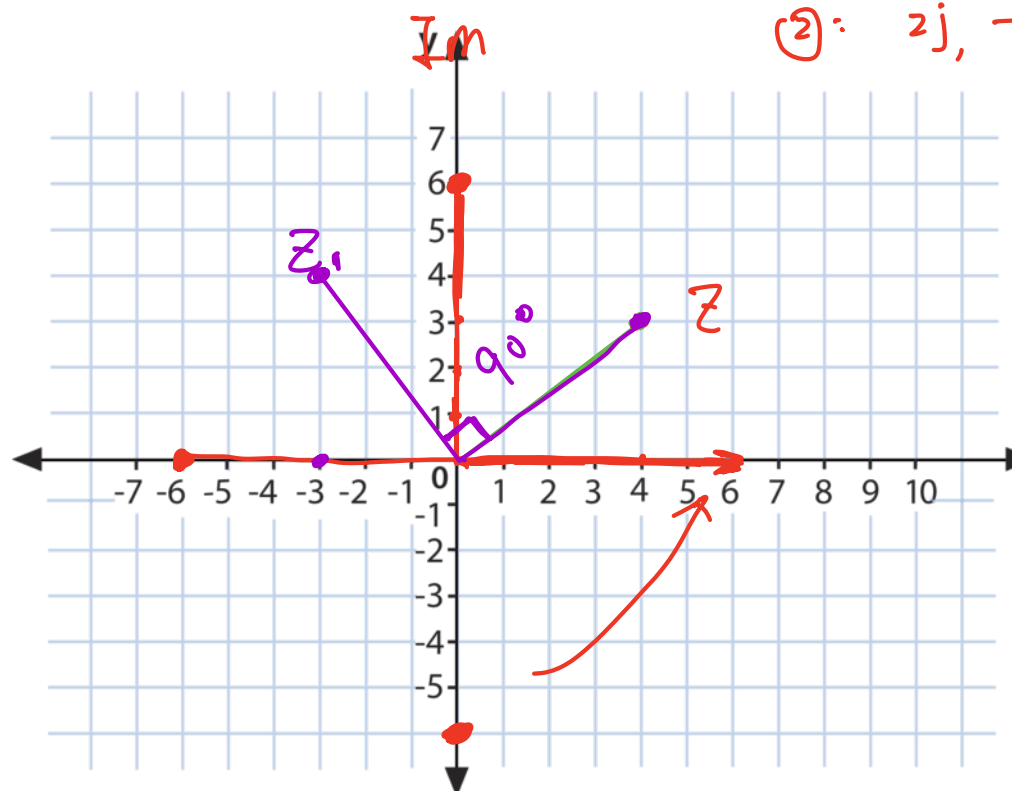


x	y	angle
+	+	$[0, 90^\circ)$
-	+	$[90^\circ, 180^\circ)$
-	-	$[180^\circ, 270^\circ)$
+	-	$[270^\circ, 360^\circ)$

1 cycle
 $\downarrow$
 360°
 $n \cdot 360^\circ$

Imaginary unit j

$$|z| = |z_1|$$



$$C = a + bi$$

①: $0j$
 ②: $zj, \rightarrow jz$

• 6
 • $j6$
 • $j(j6)$
 • $j(j^2 6)$
 $Re: j^2(j^2 6)$
 $z = 4 + j3$
 $jz = j4 + j^2 3$
 $= -3 + j4$

Conversion between forms

$$\mathbf{z}_1 = 2 + j6$$

$$\mathbf{z}_1 = 6.32 \angle 71.57^\circ$$

$$\mathbf{z}_1 = 6.32 e^{j71.57^\circ}$$

$$\mathbf{z}_2 = 2e^{j30^\circ}$$

$$\mathbf{z}_2 = 2 \angle \underline{30^\circ}$$

$$\mathbf{z}_2 = 1.732 + j$$

$$\mathbf{z}_3 = 5 \angle 275^\circ$$

$$\mathbf{z}_3 = 0.436 - j 4.981$$

$$\mathbf{z}_3 = 5 e^{j275^\circ}$$

Complex number arithmetic

Addition and subtraction

$$\underline{z_1 = x_1 + jy_1}$$

$$\underline{z_2 = x_2 + jy_2}$$

$$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$$

$$z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2).$$

$$z_1 = 2.5 + j4.6$$

$$z_2 = 5\angle 275^\circ$$

↓
convert into rect form

Multiplication

$$\mathbf{z_1 = x_1 + jy_1} \qquad \mathbf{z_2 = x_2 + jy_2}$$

Rectangular & Polar forms

$$\begin{aligned} z_1 z_2 &= (x_1 + jy_1)(x_2 + jy_2) \\ &= x_1 x_2 + j(x_1 y_2 + x_2 y_1) - y_1 y_2 \end{aligned}$$

$$z_1 = r_1 \angle \varphi_1 \quad , \quad z_2 = r_2 \angle \varphi_2$$

$$\Rightarrow z_1 z_2 = r_1 r_2 \angle (\varphi_1 + \varphi_2)$$

$$\mathbf{z_1 = 2.5 + j4.6} \qquad \mathbf{z_2 = 5 \angle 275^\circ}$$

Divisions

$$\mathbf{z_1 = x_1 + jy_1}$$

Rectangular & Polar forms

$$\mathbf{z_2 = x_2 + jy_2}$$

$$\frac{z_1}{z_2} = \frac{x_1 + jy_1}{x_2 + jy_2} = \frac{(x_1 + jy_1)(x_2 - jy_2)}{(x_2 + jy_2)(x_2 - jy_2)} = \frac{(\quad)(\quad)}{x_2^2 + y_2^2}$$

$$\mathbf{z_1 = r_1 \angle \theta_1}$$

$$\mathbf{z_2 = r_2 \angle \theta_2}$$

$$\frac{z_1}{z_2} = \frac{r_1 \angle \theta_1}{r_2 \angle \theta_2} = \frac{r_1}{r_2} \angle (\theta_1 - \theta_2)$$

$$\mathbf{z_1 = 2.5 + j4.6}$$

$$\mathbf{z_2 = 5 \angle 275^\circ}$$

Root

$$\mathbf{z}_1 = x_1 + jy_1$$

Rectangular & Polar forms

$$\sqrt{\mathbf{z}_1} \quad \times$$

$$\mathbf{z}_1 = r_1 \angle \theta_1$$

$$\sqrt{\mathbf{z}_1} = \sqrt{r_1 \angle \theta_1} = \sqrt{r_1} \angle \frac{\theta_1}{2}.$$

$$\sqrt[n]{\mathbf{z}_1} = \sqrt[n]{r_1 \angle \theta_1} = \sqrt[n]{r_1} \angle \frac{\theta_1}{n}.$$

$$\mathbf{z}_1 = 2.5 + j4.6$$

$$\mathbf{z}_2 = 5 \angle 275^\circ$$

Inverse

$$\mathbf{z_1 = x_1 + jy_1}$$

Rectangular & Polar forms

$$\mathbf{z_1 = r_1 \angle \theta_1}$$

$$\begin{aligned}\frac{1}{z_1} &= \frac{1}{x_1 + jy_1} = \frac{x_1 - jy_1}{(x_1 + jy_1)(x_1 - jy_1)} \\ &= \frac{x_1 - jy_1}{x_1^2 + y_1^2}\end{aligned}$$

$$\frac{1}{z_1} = \frac{1 \angle 0^\circ}{r_1 \angle \theta_1} = \frac{1}{r_1} \angle (0^\circ - \theta_1) = \frac{1}{r_1} \angle -\theta_1$$

$$\mathbf{z_1 = 2.5 + j4.6}$$

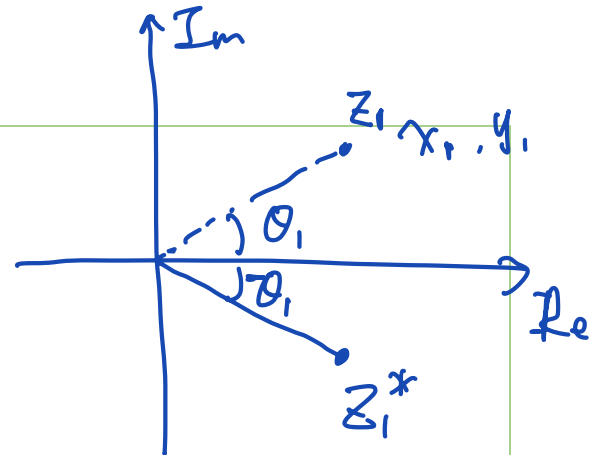
$$\mathbf{z_2 = 5 \angle 275^\circ}$$

Conjugate (*)

$$\mathbf{z_1 = x_1 + jy_1}$$

Rectangular & Polar forms

$$= x_1 - jy_1$$



$$\mathbf{z_1 = r_1 \angle \theta_1}$$

$$\mathbf{z_1^* = r_1 \angle -\theta_1}$$

$$\mathbf{z_1 = 2.5 + j4.6}$$

$$\mathbf{z_2 = 5 \angle 275^\circ}$$

Example #1: answer in three forms.

$$\mathbf{z = \frac{10 + j5 + 3\angle 40^\circ}{-3 + j4} + 10\angle 30^\circ + j5}$$